Lecture 2

Quantum states form a vector space. Their transformations are linear operators on this space, studied in linear algebra. In quantum computing, all vector spaces are over $\mathbb{C}$ (unless stated otherwise). Informally, a vector space over $\mathbb{C}$ is a set of objects that can be added among themselves and multiplied by elements of $\mathbb{C}$.

Matrices

An $n \times m$ matrix is an array of numbers:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix}$$

We write A_{ij} to denote the entry in the i-th row and j-th column of A. Above: $A_{ij} = a_{ij}$.

The set of all $n \times m$ complex matrices is denoted $M_{n,m}(\mathbb{C})$.

Basic matrix operations

$M_{n,m}(\mathbb{C})$ is a vector space itself since matrices of the same shape can be added and multiplied by a constant (scalar). Both operations are performed entry-wise: for $A, B \in M_{n,m}(\mathbb{C})$, $c \in \mathbb{C}$

- addition: $(A+B)_{ij} = A_{ij} + B_{ij}$
- scalar multiplication: $(cA)_{ij} = c A_{ij}$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & a_{22}+b_{22} \end{pmatrix} \qquad c \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} ca_{11} & ca_{12} \\ ca_{21} & ca_{22} \end{pmatrix}$$

To illustrate other basic operations, let $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$. Then

- complex conjugate: $A^* = \bar{A} = \begin{pmatrix} a_{11}^* & a_{12}^* & a_{13}^* \\ a_{21}^* & a_{22}^* & a_{23}^* \end{pmatrix}$

  $(A^*)_{ij} = (A_{ij})^*$
- transpose:

  $(A^T)_{ij} = A_{ji}$

  $A^T = \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{pmatrix}$
- conjugate transpose (adjoint):

  $(A^\dagger)_{ij} = A^*_{ji}$ $\qquad A^\dagger = (A^*)^T = (A^T)^* = \begin{pmatrix} a_{11}^* & a_{21}^* \\ a_{12}^* & a_{22}^* \\ a_{13}^* & a_{23}^* \end{pmatrix}$

Note: mathematicians often use A^* to denote $A^\dagger$.

Vectors

Vector is a matrix with only one row or column:

row vector $(a_1 \; a_2 \; \dots \; a_n)$

$\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$ column vector

Just like matrices, vectors of the same shape can be added and multiplied by scalars. They form a vector space $\mathbb{C}^n$.

Vector multiplication

A row vector can be multiplied by a column vector with the same number of entries:

$$(a_1 \; a_2 \; \dots \; a_n) \cdot \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{i=1}^{n} a_i b_i$$

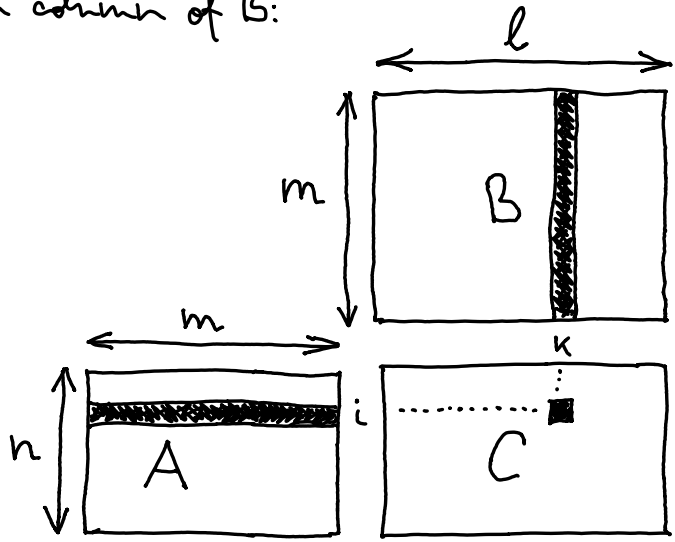
This is a special case of matrix multiplication.

Matrix multiplication

Let A be $n \times m$ and B be $m \times l$ matrices. Then their product $C = A \cdot B$ is the $n \times l$ matrix with entries

$$C_{ik} = \sum_{j=1}^{m} A_{ij} B_{jk}$$

for all $i = 1, \dots, n$ and $k = 1, \dots, l$. Note that C_{ik} is just the product of the i-th row of A and the k-th column of B:



Note: It is possible to multiply A and B only if the number of columns of A agrees with the number of rows of B:

$$[n \times m] \cdot [m \times l] = [n \times l]$$

same!

Properties of matrix multiplication

- associativity: $(AB)C = A(BC) = ABC$
- distributivity: $A(B+C) = AB + AC$
  $(A+B)C = AC + BC$
- scalar multiplication: $\lambda(AB) = (\lambda A)B = A(\lambda B) = (AB)\lambda$
- complex conjugate: $(AB)^* = A^* B^*$
- transpose: $(AB)^T = B^T A^T$ ← Reversed order!
- conjugate transpose: $(AB)^\dagger = B^\dagger A^\dagger$ ↩

Note: matrix multiplication is not commutative: $AB \neq BA$.

Dirac bra-ket notation

Column vector (ket) Row vector (bra)

Complex conjugate!

$$|\psi\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{C}^n \qquad \langle\psi| = (a_1^* \; a_2^* \; \cdots \; a_n^*)$$

Ket and bra are conjugate transposes of each other:

$$|\psi\rangle^\dagger \equiv \langle\psi| \qquad \langle\psi|^\dagger \equiv |\psi\rangle$$

For $|\psi\rangle, |\varphi\rangle \in \mathbb{C}^n$ and $A \in M_{n,n}(\mathbb{C})$ the following are valid matrix products:

- inner product: $\langle\psi| \cdot |\varphi\rangle \equiv \langle\psi|\varphi\rangle \in \mathbb{C}$
- outer product: $|\psi\rangle \cdot \langle\varphi| \equiv |\psi\rangle\langle\varphi| \in M_{n,n}(\mathbb{C})$
- matrix-vector product: $A \cdot |\psi\rangle \equiv A|\psi\rangle \in \mathbb{C}^n$
  $\langle\psi| \cdot A \equiv \langle\psi|A \in \mathbb{C}^n$ (a row vector)

Inner product

The inner product of vectors $|a\rangle, |b\rangle \in \mathbb{C}^n$ is the complex number $\langle a|b\rangle$.

If $|a\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}$ and $|b\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix}$ then

$$\langle a|b\rangle = (\alpha_1^* \; \alpha_2^* \; \cdots \; \alpha_n^*) \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} = \sum_{i=1}^{n} \alpha_i^* \beta_i \in \mathbb{C}$$

Note:

- $\langle a|b\rangle = (\langle b|a\rangle)^+ = \overline{\langle b|a\rangle}$
- $\langle a|a\rangle = \sum_{i=1}^{n} |a_i|^2$ is a real number
- $\langle a|a\rangle \geq 0$ and $\langle a|a\rangle = 0 \iff |a\rangle = 0$ (the all-zeroes vector)

Norm and orthogonality

The norm (or "length") of $|a\rangle \in \mathbb{C}^n$ is

$$\||a\rangle\| := \sqrt{\langle a|a\rangle} = \sqrt{\sum_{i=1}^{n} |\alpha_i|^2} \geq 0$$

A unit vector has norm 1: $\sum_{i=1}^{n} |\alpha_i|^2 = 1$.

Recall: qubit states $\alpha|0\rangle + \beta|1\rangle$ are unit vectors since $|\alpha|^2 + |\beta|^2 = 1$.

Cauchy – Schwarz inequality:

$$|\langle a|b\rangle| \leq \||a\rangle\| \cdot \||b\rangle\| \quad \text{for any } |a\rangle, |b\rangle \in \mathbb{C}^n$$

If $|a\rangle$ and $|b\rangle$ are unit vectors then

$|\langle a|b\rangle| \in [0,1]$ so $|\langle a|b\rangle| = \cos\theta \in [0,1]$ for some angle $\theta \in [0, \pi/2]$. The vectors are orthogonal if $\langle a|b\rangle = 0$, i.e., $\theta = \frac{\pi}{2}$.

Note: when restricted to real vectors, the notions of norm and orthogonality coincide with what we are used to. The only difference is that complex vectors cannot have obtuse (larger than 90° or $\pi/2$) angles.

Outer product and projector

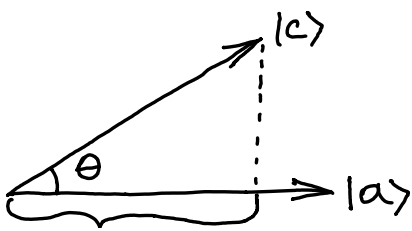
For any $|a\rangle \in \mathbb{C}^n$ and $|b\rangle \in \mathbb{C}^m$, the matrix $|a\rangle\langle b| \in M_{n,m}(\mathbb{C})$ is called the <u>outer product</u> of $|a\rangle$ and $|b\rangle$. It satisfies, for any $|c\rangle \in \mathbb{C}^m$,

$$\big(|a\rangle\langle b|\big)\cdot|c\rangle = \langle b|c\rangle \cdot |a\rangle$$

If $|a\rangle$ is a unit vector then $|a\rangle\langle a|$ is called the <u>projector</u> onto the one-dimensional subspace generated by $|a\rangle$. This is particularly clear for real vectors: if $|a\rangle, |c\rangle \in \mathbb{R}^n$ are real and we set $\cos\theta = \langle a|c\rangle$ then

$$|a\rangle\langle a|c\rangle = \langle a|c\rangle\, |a\rangle = \cos\theta\, |a\rangle$$

Here is a graphical depiction of this:



$\cos\theta\,|a\rangle$ ← the projection of $|c\rangle$ on the unit vector $|a\rangle$

Basis

A basis of $\mathbb{C}^n$ is a minimal collection of vectors $|b_1\rangle, \ldots, |b_n\rangle$ such that any vector $|a\rangle \in \mathbb{C}^n$ can be expressed as a linear combination of them:

$$|a\rangle = \alpha_1 |b_1\rangle + \cdots + \alpha_n |b_n\rangle$$

for some coefficients $\alpha_i \in \mathbb{C}$. Minimality implies that $|b_1\rangle, \ldots, |b_n\rangle$ are linearly independent, meaning that no $|b_i\rangle$ can be expressed as a linear combination of the rest. The size of a basis is uniquely determined by the vector space and is called its dimension.

Example: The dimension of $\mathbb{C}^n$ is n.
The dimension of the set of all $n \times m$ complex matrices $M_{n,m}(\mathbb{C})$ as a linear space is $n \cdot m$.

For a given basis $|b_1\rangle, \ldots, |b_n\rangle \in \mathbb{C}^n$, every vector $|a\rangle \in \mathbb{C}$ can be uniquely represented by the n-tuple $\alpha_1, \ldots, \alpha_n$ above called its coordinates.

An orthonormal basis consists of pair-wise orthogonal unit vectors $|b_1\rangle, \ldots, |b_n\rangle$, meaning

$$\langle b_i | b_j \rangle = \delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Standard basis

Here are three different bases for $\mathbb{C}^2$:

$$\begin{pmatrix}3\\2\end{pmatrix}, \begin{pmatrix}4\\-i\end{pmatrix} \qquad \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix} \qquad \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

The last two are orthonormal. The <u>standard</u> or <u>computational</u> basis for $\mathbb{C}^n$ is

$$|1\rangle \equiv \begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix} \quad |2\rangle \equiv \begin{pmatrix}0\\1\\\vdots\\0\end{pmatrix} \quad \cdots \quad |n\rangle \equiv \begin{pmatrix}0\\0\\\vdots\\1\end{pmatrix}$$

(Sometimes we will start from $|0\rangle$, e.g., when $n=2$.)

Any vector $|a\rangle \in \mathbb{C}^n$ can be expanded in the standard basis:

$$|a\rangle = \begin{pmatrix}\alpha_1\\\vdots\\\alpha_n\end{pmatrix} = \sum_{i=1}^{n} \alpha_i |i\rangle$$

You can similarly expand a matrix. Notice that

$$|1\rangle\langle 2| = \begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix} \cdot (0\ 1\ \cdots\ 0) = \begin{pmatrix}0 & 1 & \cdots & 0\\0 & 0 & \cdots & 0\\\vdots & \vdots & \ddots & \vdots\\0 & 0 & \cdots & 0\end{pmatrix}$$

(row 1 → ; column 2 ↓)

Outer products form the <u>standard basis for matrices</u>:

$$\begin{pmatrix}a_{11} & a_{12} & \cdots & a_{1m}\\a_{21} & a_{22} & \cdots & a_{2m}\\\vdots & \vdots & \ddots & \vdots\\a_{n1} & a_{n2} & \cdots & a_{nm}\end{pmatrix} = \sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij} |i\rangle\langle j|$$

($|i\rangle\langle j|$: 1 at coordinates (i,j) and zeroes everywhere else; $|i\rangle$: n entries; $\langle j|$: m entries)

Recovering the (i,j)-th entry of A: $\langle i|A|j\rangle = a_{ij}$.

Eigenvalues and eigenvectors

An eigenvector of a square matrix $A \in M_{n,n}(\mathbb{C})$ is a non-zero vector $|v\rangle \in \mathbb{C}^n$ such that

$$A|v\rangle = \lambda|v\rangle$$

for some $\lambda \in \mathbb{C}$ called the eigenvalue corresponding to $|v\rangle$.
The eigenvalues of A are the roots of the characteristic polynomial $\det(A - \lambda I) = 0$
where det denote the determinant and

$$I = \sum_{i=1}^{n} |i\rangle\langle i| = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

is the $n \times n$ identity matrix.
Each $n \times n$ matrix has at least one eigenvalue.

Diagonal representation

A square matrix $A \in M_{n,n}(\mathbb{C})$ is diagonalisable if

$$A = \sum_{i=1}^{n} \lambda_i |v_i\rangle\langle v_i|$$

where $|v_1\rangle, \ldots, |v_n\rangle \in \mathbb{C}^n$ is an orthonormal set of eigenvectors of A with eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$.
Note that $A|v_j\rangle = \sum_{i=1}^{n} \lambda_i |v_i\rangle\underbrace{\langle v_i|v_j\rangle}_{\delta_{ij}} = \lambda_j |v_j\rangle$, meaning that A is diagonal

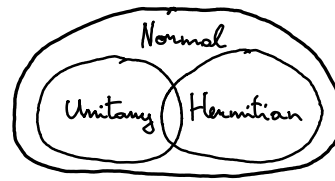
$$\begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$$

in the basis of its eigenvectors.

Normal, Hermitian and unitary matrices

A square matrix $A \in M_{n,n}(\mathbb{C})$ is <u>normal</u> if

$$A^\dagger A = A A^\dagger$$



<u>Theorem</u> (p. 72 of Nielsen & Chuang)
A matrix is diagonalizable if and only if it is normal.

A square matrix $A \in M_{n,n}(\mathbb{C})$ is <u>Hermitian</u> if

$$A = A^\dagger$$

Such matrix is clearly normal and hence diagonalisable.
A normal matrix is Hermitian iff it has real eigenvalues.

A square matrix $A \in M_{n,n}(\mathbb{C})$ is <u>unitary</u> if

$$A A^\dagger = A^\dagger A = I$$

Recall from Lecture 1:

$$\begin{pmatrix} \bar{a} & \bar{b} \\ \bar{c} & \bar{d} \end{pmatrix} \cdot \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} = \begin{pmatrix} |a|^2+|b|^2 & \bar{a}c+\bar{b}d \\ a\bar{c}+b\bar{d} & |c|^2+|d|^2 \end{pmatrix}$$

where I is the $n \times n$ identity matrix. Unitary matrices are normal and therefore diagonalisable. All eigenvalues of a unitary operator have absolute value 1. Unitary operators preserve inner products: if U is unitary and

$$\left.\begin{aligned} |a'\rangle &= U|a\rangle \\ |b'\rangle &= U|b\rangle \end{aligned}\right\} \Rightarrow \langle a'|b'\rangle = (U|a\rangle)^\dagger (U|b\rangle) = \langle a|U^\dagger U|b\rangle = \langle a|b\rangle$$

Any unitary matrix U is of the form $U = e^{iH}$ for some Hermitian matrix H. You can think of Hermitian matrices as real numbers and unitary matrices as complex numbers of the form $e^{i\varphi}$ for some $\varphi \in \mathbb{R}$.

Tensor product

Let A and B be matrices of arbitrary shape. Their tensor product (or Kronecker product) is the following block matrix:

$$A \otimes B = \begin{pmatrix} A_{11}B & A_{12}B & \cdots & A_{1m}B \\ A_{21}B & A_{22}B & \cdots & A_{2m}B \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1}B & A_{n2}B & \cdots & A_{nm}B \end{pmatrix}$$

where the block at coordinates (i,j) is equal to $A_{ij}B$. If A is $n \times m$ and B is $n' \times m'$ then $A \otimes B$ is $nn' \times mm'$.

Tensor product applies also to vectors: $|\psi\rangle \otimes |\varphi\rangle$ and $\langle\psi| \otimes \langle\varphi|$.
Note: We abbreviate these as $|\psi\rangle|\varphi\rangle$ and $\langle\psi|\langle\varphi|$.

You can also mix vectors and matrices.
For example: $\langle 0| \otimes I$ and $I \otimes |\psi\rangle$.

Some elementary properties of the tensor product:

- associativity: $(A \otimes B) \otimes C = A \otimes (B \otimes C) = A \otimes B \otimes C$
- distributivity: $A \otimes (B+C) = A \otimes B + A \otimes C$
  $(A+B) \otimes C = A \otimes C + B \otimes C$
- scalar multiplication: $\lambda(A \otimes B) = (\lambda A) \otimes B = A \otimes (\lambda B)$

Tensor product interacts nicely with matrix multiplication. Whenever the matrix sizes are such that A_1A_2 and B_1B_2 make sense, then

$$(A_1 \otimes B_1) \cdot (A_2 \otimes B_2) = (A_1A_2) \otimes (B_1B_2).$$